

**Research  
Article**

# Aeroelastic Stability Analysis of Wind Turbines Using an Eigenvalue Approach

M. H. Hansen, Wind Energy Department, Risø National Laboratory, P.O. Box 49, DK-4000 Roskilde, Denmark

**Key words:**  
horizontal axis  
turbines;  
mechanical  
vibrations;  
dynamic stability

*A design tool for performing aeroelastic stability analysis of wind turbines is presented in this paper. The method behind this tool is described in a general form, as independent of the particular aeroelastic modelling as possible. Here, the structure is modelled by a Finite beam Element Method, and the aerodynamic loads are modelled by the Blade Element Momentum method coupled with a Beddoes-Leishman type dynamic stall model in a state-space formulation. The linearization of the equations of motion is performed about a steady-state equilibrium, where the deterministic forcing of the turbine is neglected. To eliminate the periodic coefficients and avoid using the Floquet Theory, the multi-blade transformation is utilized. From the corresponding eigenvalue problem, the eigenvalues and eigenvectors can be computed at any operation condition to give the aeroelastic modal properties: Natural frequencies, damping and mode shapes. An example shows a good agreement between predicted and measured aeroelastic damping of a stall-regulated 600 kW turbine. Copyright © 2004 John Wiley & Sons, Ltd.*

## Introduction

A new design tool for performing aeroelastic stability analysis of wind turbines is presented. The tool may be the first of its kind that is based on eigenvalue analysis and includes dynamic stall.

The presented work is part of the contribution by Risø National Laboratory to the STABCON project (Aeroelastic Stability and Control of Large Wind Turbines) funded by the European Commission. This project aims at improving the tools for predicting the aeroelastic stability (damping) of turbines, and the tools for design of control systems. The latter has become an important objective in the continuous optimization of wind turbines with active power regulation. The next generation of these turbines will be fitted with control systems that, beside power optimization, also actively alleviate loads and actively suppress aeroelastic instabilities under critical operation conditions. Optimal designs of such control systems depend on a linear aeroelastic stability tool.

An early example of such a linear stability tool called ARLIS was represented by Kirchgäßner [1] at EWEC 1984. The tool was originally developed for rotorcraft applications, but can also be used for analyzing wind turbines, as Kirchgäßner showed for an experimental two-bladed machine. Using Floquet Theory for the stability analysis of turbines, ARLIS was ahead of its time, even though the aerodynamic modelling is based on a quasi-steady assumption.

Later as the first stall-induced vibrations were observed on commercial wind turbines in the early 90's [2], several tools became available for analyzing stability of wind turbines, besides the aeroelastic codes for con-

\* Correspondence to: M. H. Hansen, Wind Energy Department, Risø National Laboratory, PO Box 49, Dk-4000 Roskilde, Denmark  
E-mail: morten.hansen@risoe.dk  
Contract/grant sponsor: European Commission; Contract/grant number: ENK5-CT-2002-00627.

ducting traditional time-domain analysis. Most of these tools are based on the same method as used in the European STALLVib project, which was aimed at the stall-induced vibrations [3]. In this method, the aerodynamic damping of a blade vibration is approximated as the work done by the aerodynamic forces on the blade over one period of oscillation. The advantage of this method is that the aerodynamic forces may be computed from a nonlinear model, whereby the amplitude dependency of the aerodynamic damping can be investigated. The nonlinear effects of some dynamic stall models have been shown to limit the predicted aerodynamic damping as the amplitude increases [3, 4], which is important when the loads on the wind turbine are estimated. The disadvantage of approximating damping from work computations is that the prescribed blade vibration must be known on forehand. Often the structural mode shapes of the blades are used in stability assessments with these tools, based on the assumption that the aeroelastic and structural mode shapes are identical. This is not the case, especially for a classical flutter instability. Furthermore, it has been shown in [5] that blades do not vibrate in pure blade modes when they operate on the rotor; flapwise and edgewise bending modes may couple in the rotor whirling modes of the turbine.

Recently, new stability tools for blade only analysis have been developed by different partners of the European DAMPBLADE project [6]. Since there are no periodic coefficients in the aeroelastic equations of motion for a blade rotating without influence from the remaining turbine, these tools are based on eigenvalue analysis, similar to the tool developed by Chaviaropoulos [7]. The main issues in developing the new tools have been the linearization and state-space formulation of the aerodynamic models, which normally are used in nonlinear indicial formulations of existing aeroelastic codes. These issues have been solved, and the solutions form the basis in new stability tools for predicting the modal aeroelastic damping of the entire turbine.

A new stability tool called *HAWCStab* for turbine analysis developed at Risø is presented in this paper. The paper focuses on the concepts behind the tool, and not so much on the detailed modelling. The basic model is similar to that of the aeroelastic code HAWC [8, 9]: The structure is modelled by a Finite beam Element Method, and the aerodynamic loads are modelled by the Blade Element Momentum method coupled with a Beddoes-Leishman type dynamic stall model. This model of unsteady aerodynamics is formulated in state-space by a set of ordinary differential equations [10]. The linearization of the equations of motion is performed about a steady-state equilibrium, where the deterministic forcing of the turbine is neglected. To eliminate the periodic coefficients in the resulting linear equations of motion and avoid using the Floquet Theory, the multi-blade transformation is utilized. From the corresponding eigenvalue problem, the eigenvalues and eigenvectors can be computed at any operation condition to give the aeroelastic modal properties: Natural frequencies, damping and mode shapes. This approach is limited to symmetric rotors (identical blades with the same pitch) because of the multi-blade transformation. However the stability tool can in principle handle ground fixed asymmetries, e.g. gravity loads, rotor tilt, and yaw errors; although the present version of *HAWCStab* still is limited to symmetric inflow and no gravity.

The paper is structured as follows: First, the linear aeroelastic model is described in general terms with focus on the issues that are important for setting up the eigenvalue problem. Second, the eigenvalue analysis is described including issues about conditioning the eigenvalue problem and interpretation of the eigensolutions. Finally, a stability analysis is presented for a stall-regulated 600 kW turbine, which has been used in an experiment to estimate the aeroelastic modal damping of its edgewise whirling modes [11].

## Linear Aeroelastic Model

The linear aeroelastic model of three-bladed horizontal axis wind turbines used in *HAWCStab* is described in this section. The turbine is modelled by articulated Timoshenko beam elements using the Finite Element Method, where the aerodynamic forces obtained by a Blade Element Momentum method are transformed to the nodes assuming parabolic distribution over the blade elements. The linear equations of motion are derived using Lagrange's equations in a linear formulation. To enable the eigenvalue analysis, the periodic coefficients in these equations are eliminated by formulating the structural and aerodynamic degrees of freedom (DOFs) for the rotor in *multi-blade coordinates*.

## Multi-Blade Coordinates

The multi-blade transformation, or Coleman transformation for bladed rotors [12, 13] is a method to describe the individual blade coordinates in the ground fixed frame of reference. The periodic terms in governing aeroelastic equations can thereby be eliminated when the number of rotor blades is odd, the rotor is *isotropic* (identical and symmetrically mounted blades), and the inflow to the rotor is uniform. These conditions can be expanded to ground fixed cyclic variations in blade properties and inflow, e.g. due to tilt and yaw of rotor, or cyclic blade pitch.

The periodic terms are eliminated because all coordinates of the model are defined in the same frame of reference. The wind field and the motion of the tower and nacelle are defined in the ground fixed frame. The bending of the main shaft is defined in the ground fixed frame by a simple rotational transformation of the shaft coordinates with respect to the azimuth angle. A physical coordinate  $q_k$  for blade number  $k$ , defined in its own co-rotating frame, is transformed into the ground fixed frame by the multi-blade transformation:

$$q_k(t) = a_0(t) + a_1(t) \cos\left(\Omega t + \frac{2\pi}{3}(k-1)\right) + b_1(t) \sin\left(\Omega t + \frac{2\pi}{3}(k-1)\right) \quad (1)$$

where  $t$  is time,  $k = 1, 2, 3$  is the blade number, and  $\Omega$  is the mean rotational speed of the rotor. Three multi-blade coordinates  $a_0$ ,  $a_1$ , and  $b_1$  replace the three blade coordinates  $q_1$ ,  $q_2$ , and  $q_3$ . To understand that the multi-blade coordinates describe the rotor coordinates in the ground fixed frame, assume that blade coordinates  $q_k$  describes a flapwise blade deflection. In that case, the  $a_0$  coordinate describes a simultaneous flapwise deflection of all three blades, while the  $a_1$  and  $b_1$  coordinates describe tilt and yaw motions, respectively.

Similar considerations can be used for the remaining blade coordinates of both structural and aerodynamic DOFs in an aeroelastic model. The following two sections describes the linear aeroelastic model used for the eigenvalue analysis in HAWCStab, where all blade coordinates are defined in multi-blade coordinates by the transformation (1).

## Structural Dynamics

The turbine structure is modelled by articulated prismatic beam elements as shown schematically in Figure 1. The beam structures can be divided into three different substructures: Tower, nacelle including the shaft, and blades. These substructures are kinematically coupled at the tower top and rotor center nodes.

The tower is assumed to be clamped at the first node of the tower substructure. The tower top node has the same DOFs as the first node of nacelle substructure, except for a *constant* speed azimuthal rotation of the shaft node. In this first version of HAWCStab there is no rotational DOF for modelling the generator. The nacelle substructure models both nacelle deformations and torsional deformations of the drive-train, which is rotating at the constant rotor speed. The three blades are connected to the nacelle at the rotor center, whereby the first common node of the blades has the same DOFs as the shaft end, when accounting for the 120° azimuthal blade positions.

The Timoshenko beam theory, used for the prismatic beam elements, includes the shear deformations and rotational inertia of the beam cross-sections. Each node has therefore six DOFs: Two pairs of cross-sectional translations and rotations, a longitudinal translation, and a torsional rotation. The shape functions used to describe the variations of these deformations over the element as function of the element coordinate and node DOFs are similar to the functions suggested in [14]. The six deformations of all beam cross-sections of the turbine model is thereby described by the node deformation vector (defined in the ground fixed frame with multi-blade coordinates):

$$\mathbf{u} = \begin{Bmatrix} \mathbf{u}_t \\ \mathbf{u}_n \\ \mathbf{u}_{a_0} \\ \mathbf{u}_{a_1} \\ \mathbf{u}_{b_1} \end{Bmatrix} \quad (2)$$

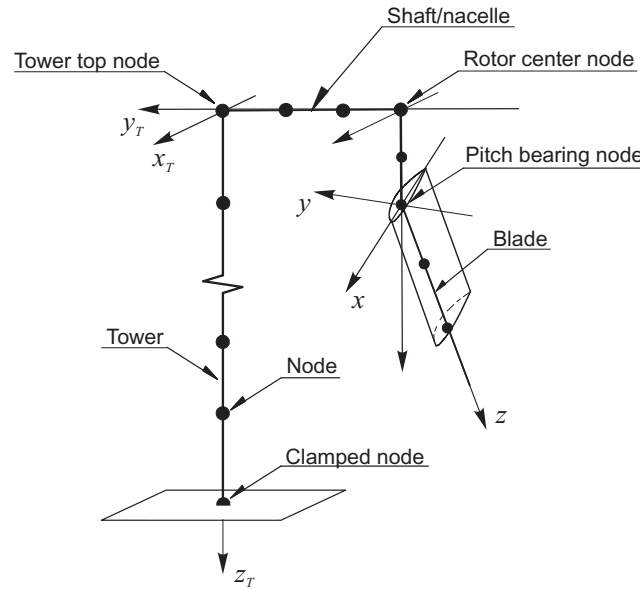


Figure 1. Schematics of the articulated beam element model of a wind turbine used in HAWCStab. Note that only one blade is drawn

where  $\mathbf{u}_t$  contains the tower substructure node DOFs,  $\mathbf{u}_n$  contains the nacelle substructure node DOFs described in the fixed frame of reference,  $\mathbf{u}_{a_0}$  contains the symmetric node DOFs of the rotor blades, and the two vectors  $\mathbf{u}_{a_1}$  and  $\mathbf{u}_{b_1}$  contain the two asymmetric node DOFs of the rotor blades (cf. equation (1)).

The equations of motion are derived using Lagrange's equation. The total kinetic energy can be written as

$$T = \sum_{k=1}^{N_e} T_k(\mathbf{u}, \dot{\mathbf{u}}) \quad (3)$$

where  $N_e$  is the total number of elements,  $\dot{\mathbf{u}}$  is the node velocity vector, and  $T_k$  is the kinetic energy of element number  $k$ , which is derived as

$$T_k(\mathbf{u}, \dot{\mathbf{u}}) = \frac{1}{2} \frac{m_k}{A_k} \int_0^{l_k} \int_A \dot{\mathbf{r}}_k^T \dot{\mathbf{r}}_k dA dz \quad (4)$$

where  $\dot{\mathbf{r}}_k = \dot{\mathbf{r}}_k(x, y, z; \mathbf{u}, \dot{\mathbf{u}})$  is a velocity vector in the ground fixed frame for the cross-sectional mass particle with mass  $m_k/A_k$  at cross-sectional position  $(x, y)$  and element coordinate  $z$ , and  $l_k$ ,  $m_k$  and  $A_k$  are the length, mass per unit-length and area of the cross-section for the prismatic element  $k$ . This velocity vector is obtained as the time derivative of the position vector, which is not written out here for the brevity of this paper.

The total potential energy can be written as

$$V = \sum_{k=1}^{N_e} V_k(\mathbf{u}) \quad (5)$$

where  $V_k$  is the potential energy stored in element number  $k$ , which for rotor elements has the form

$$V_k(\mathbf{u}) = V_{e,k}(\mathbf{u}) + V_{c,k}(\mathbf{u}) \quad (6)$$

where  $V_{e,k}$  is the elastic strain energy, and  $V_{c,k}$  is the potential energy due to centrifugal blade forces. The geometric nonlinearity that an element is elongated by variations in cross-sectional deformations, causes energy from the centrifugal normal forces to be stored in the blade elements.

It is important to note that the presumption of an isotropic rotor for the use of multi-blade coordinates to eliminate the periodic coefficients in the equations of motion, also implies that the blade nodes are identical for all three rotor blades. Each set of three identical blade elements can therefore be collected to a single *rotor element*, which simplifies the derivation of the equation of motion using Lagrange's equation on element level. The contribution to the Lagrangian  $L = T - V$  from a rotor element is the sum of the contributions from the three identical blade elements:

$$L_k^r = T_k^{B_1} + T_k^{B_2} + T_k^{B_3} - V_k^{B_1} - V_k^{B_2} - V_k^{B_3} \quad (7)$$

where the subscript here refers to rotor element number  $k$ . Similar to the contributions from a tower element  $L_k^t$  and a nacelle element  $L_k^n$ , this collected contribution from a rotor element is independent of time  $t$  because of the multi-blade formulation. The element contributions to the Lagrangian depend only on the structural state  $L_k = L_k(\mathbf{u}, \dot{\mathbf{u}})$ , where the superscripts are omitted.

Lagrange equations including generalized aerodynamic forces,  $Q_i$ , can be written as

$$\sum_k \left( \frac{\partial}{\partial t} \left( \frac{\partial L_k}{\partial \dot{u}_i} \right) - \frac{\partial L_k}{\partial u_i} \right) = Q_i, \quad i = 1, 2, \dots, N \quad (8)$$

where the sum over index  $k$  includes the tower, nacelle, and rotor elements, and  $N = 6 \times N_e$  is the total number of generalized coordinates (DOFs). Note that there is no dissipation function introduced here; structural damping will be added to the equation of motion as element matrices based on a Rayleigh type damping model [15].

Linearization of Lagrange's equations (8) and introduction of structural damping lead to the following linear equation of motion

$$\mathbf{M}\ddot{\mathbf{u}} + (\mathbf{C} + \mathbf{G})\dot{\mathbf{u}} + \mathbf{K}\mathbf{u} = \mathbf{Q} \quad (9)$$

where the damping matrix  $\mathbf{C}$  is derived from the Rayleigh type damping model. Because the contributions to the Lagrangian from all elements are autonomous, the contributions of element  $k$  to the mass, gyroscopic, and stiffness matrices can be derived as

$$\begin{aligned} m_{ij,k} &= \left. \frac{\partial^2 L_k}{\partial \dot{u}_i \partial \dot{u}_j} \right|_{\mathbf{u}=\mathbf{u}_0, \dot{\mathbf{u}}=0} \\ g_{ij,k} &= \left. \left( \frac{\partial^2 L_k}{\partial \dot{u}_i \partial u_j} - \frac{\partial^2 L_k}{\partial u_i \partial \dot{u}_j} \right) \right|_{\mathbf{u}=\mathbf{u}_0, \dot{\mathbf{u}}=0} \\ k_{ij,k} &= - \left. \frac{\partial^2 L_k}{\partial u_i \partial u_j} \right|_{\mathbf{u}=\mathbf{u}_0, \dot{\mathbf{u}}=0} \end{aligned} \quad (10)$$

where  $\mathbf{u}_0$  is the steady state deformation of the turbine, which in this first version of HAWCStab is assumed to be a symmetric deformation of the rotor. Note that the mass and stiffness matrices are symmetric, and the gyroscopic matrix is anti-symmetric.

## Aerodynamics

The distribution of aerodynamic forces over an blade element is assumed to be parabolic, whereby three aerodynamic calculation points are needed for each element. To reduce the total number of calculation points and ensure that the distribution over the entire blade is continuous, the calculation points are placed at the nodes and the middle of each element. Because the aerodynamic forces are assumed to vanish at the tip and rotor center, the number of aerodynamic calculation points is  $N_{ap} = 2N_{eb} - 1$  for each blade, where  $N_{eb}$  is the number of blade elements.

A Beddoes-Leishman type dynamic stall model in a state-space formulation recently developed at Risø [10] is used to model the unsteady forces in each aerodynamic calculation point. The model includes the dynamic effects of the near wake (Theodorsen Theory) and the trailing edge separation in stall on the lift, drag, and moment coefficients. However, the effect of leading edge separation is neglected because of the moderate reduced frequencies and types of airfoils applicable for turbine blades. Added fluid mass is also neglected due to the moderate reduced frequencies.

Figure 2 shows an example of typical loops of the unsteady lift and drag coefficients for vibrations about different angles of attack. The model of the trailing separation is highly nonlinear, however it can be linearized with reasonable agreement between the linear and nonlinear loops, even for relatively large variations in angle of attack, as seen in Figure 2.

The model uses four aerodynamic states to describe the dynamics of the unsteady lift, drag, and moment coefficients [10]: Two states to model the two time-lags in the near wake effect (corresponding to a two time-lag approximation to the indicial function of the Theodorsen theory), and two time-lags in the effect of trailing edge separation (one lag in the pressure distribution over the chord, and one lag in movement in the separation point). Hence, the total number of aerodynamic states is  $N_a = 4 \times 3 \times N_{ap}$ , which are collected in an aerodynamic state vector:

$$\mathbf{x} = \begin{Bmatrix} \mathbf{x}_{a_0} \\ \mathbf{x}_{a_l} \\ \mathbf{x}_{b_l} \end{Bmatrix} \quad (11)$$

where  $\mathbf{x}_{a_0}$  contains the symmetric aerodynamic states, and the two vectors  $\mathbf{x}_{a_l}$  and  $\mathbf{x}_{b_l}$  contain the two asymmetric aerodynamic states of the rotor blades, as defined by equation (1).

The linear first order dynamic equations for the aerodynamic states can be written as [10]

$$\dot{\mathbf{x}} + \mathbf{A}_d \mathbf{x} = \mathbf{H}(\mathbf{u}, \dot{\mathbf{u}}) \quad (12)$$

where  $\mathbf{A}_d$  is a matrix having uncoupled  $4 \times 4$  matrices in the diagonal, one for each aerodynamic calculation point. The vector on the right hand side is an inherent nonlinear function of the deformation and velocity vectors describing the effect of turbine vibrations on the angle of attack, pitch rates, and relative inflow velocity at the calculation points. This function is linearized as

$$\mathbf{H}(\mathbf{u}, \dot{\mathbf{u}}) \approx \mathbf{C}_{ux} \dot{\mathbf{u}} + \mathbf{K}_{ux} \mathbf{u} \quad (13)$$

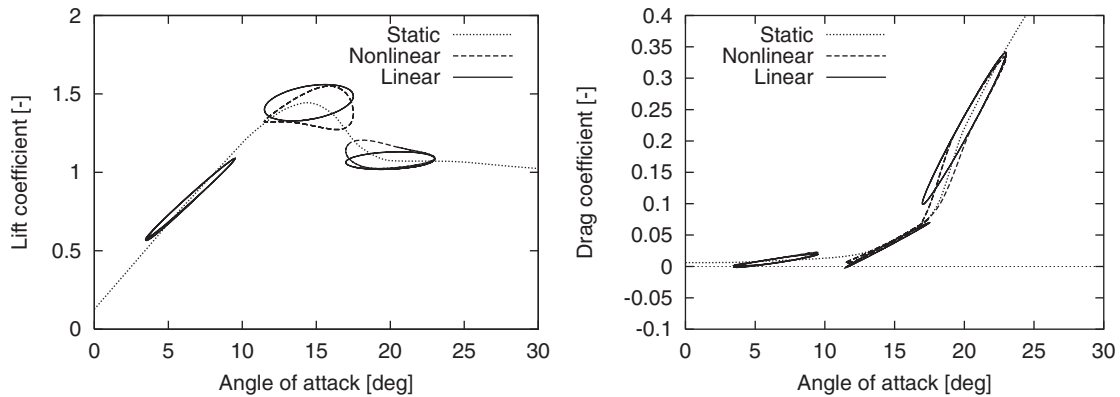


Figure 2. Linear and nonlinear unsteady lift and drag coefficients computed from the Beddoes-Leishman type dynamic stall model developed in a state-space formulation by Hansen et al. [10]. Model parameters and reduced frequencies used in this example are typical for turbine blades



The linearization of the dynamic stall model to obtain the matrix  $\mathbf{A}_d$  and linear approximation of the function  $\mathbf{H}$  is performed about the steady state given by the symmetric deformation  $\mathbf{u}_0$  of the rotor and a symmetric aerodynamic state  $\mathbf{x}_0$ . This steady state is computed simultaneously with the induced velocities to obtain the influence of the fixed wake on the steady state conditions.

The aerodynamic forces and moments in the calculation points on the blades are given by nonlinear functions of the aerodynamic states  $\mathbf{x}$ , and the structural states  $\mathbf{u}$  and  $\dot{\mathbf{u}}$  due to their effect on the angles of attack, pitch rates, and relative inflow velocities. The three calculation points for each blade element enable the derivation of a parabolic distribution of cross-sectional forces and torsional moment over the element  $k$ :

$$\mathbf{F}_k = \mathbf{F}_k(z; \mathbf{u}, \dot{\mathbf{u}}, \mathbf{x}) \quad \text{and} \quad M_k = M_k(z; \mathbf{u}, \dot{\mathbf{u}}, \mathbf{x}) \quad (14)$$

where  $z$  is the length coordinate of the element. These forces (which are not written out here for the brevity) are defined in the ground fixed frame, and they are applied in the aerodynamic center of the blade cross-section, resulting in the generalized forces [16]:

$$Q_i = \sum_k \int_0^{l_k} \left( \mathbf{F}_k^T \frac{\partial \mathbf{r}_{AC,k}}{\partial u_i} + M_k \frac{\partial \theta_{AC,k}}{\partial u_i} \right) dz, \quad i = 1, 2, \dots, N \quad (15)$$

where the sum over index  $k$  includes all blade elements, the vector  $\mathbf{r}_{AC,k} = \mathbf{r}_{AC,k}(z; \mathbf{u})$  describes the position of the aerodynamic center in the ground fixed frame, and  $\theta_{AC,k} = \theta_{AC,k}(z; \mathbf{u})$  is the torsion of the cross-section.

After linearization of (15) about the steady state  $(\mathbf{u}_0, \mathbf{x}_0)$  the generalized aerodynamic forces can be approximated by

$$\mathbf{Q}(\mathbf{u}, \dot{\mathbf{u}}, \mathbf{x}) \approx \mathbf{C}_a \dot{\mathbf{u}} + \mathbf{K}_a \mathbf{u} + \mathbf{A}_f \mathbf{x} \quad (16)$$

where the contribution to the aerodynamic damping and stiffness matrices ( $\mathbf{C}_a$  and  $\mathbf{K}_a$ ) and the coupling matrix  $\mathbf{A}_f$  from each blade element can be derived from

$$\begin{aligned} c_{a,ij,k} &= \int_0^{l_k} \left( \frac{\partial \mathbf{F}_k^T}{\partial \dot{u}_j} \frac{\partial \mathbf{r}_{AC,k}}{\partial u_i} + \frac{\partial M_k}{\partial \dot{u}_j} \frac{\partial \theta_{AC,k}}{\partial u_i} \right) \bigg|_{\substack{\mathbf{u}=\mathbf{u}_0 \\ \dot{\mathbf{u}}=\mathbf{0} \\ \mathbf{x}=\mathbf{x}_0}} dz \\ k_{a,ij,k} &= \int_0^{l_k} \left( \frac{\partial \mathbf{F}_k^T}{\partial u_j} \frac{\partial \mathbf{r}_{AC,k}}{\partial u_i} + \frac{\partial M_k}{\partial u_j} \frac{\partial \theta_{AC,k}}{\partial u_i} + \mathbf{F}_{0,k}^T \frac{\partial \mathbf{r}_{AC,k}}{\partial u_i \partial u_j} + M_{0,k} \frac{\partial \theta_{AC,k}}{\partial u_i \partial u_j} \right) \bigg|_{\substack{\mathbf{u}=\mathbf{u}_0 \\ \dot{\mathbf{u}}=\mathbf{0} \\ \mathbf{x}=\mathbf{x}_0}} dz \\ a_{f,ij,k} &= \int_0^{l_k} \left( \frac{\partial \mathbf{F}_k^T}{\partial x_j} \frac{\partial \mathbf{r}_{AC,k}}{\partial u_i} + \frac{\partial M_k}{\partial x_j} \frac{\partial \theta_{AC,k}}{\partial u_i} \right) \bigg|_{\substack{\mathbf{u}=\mathbf{u}_0 \\ \dot{\mathbf{u}}=\mathbf{0} \\ \mathbf{x}=\mathbf{x}_0}} dz \end{aligned} \quad (17)$$

where  $\mathbf{F}_{0,k}$  and  $M_{0,k}$  are the steady state force and moment distributions for element  $k$ . These matrix contributions do not have any type of symmetry, which corresponds to the fact that the aerodynamic forces  $\mathbf{Q}$  are circulatory.

Collection of equations (9), (12), (13), and (16) yields the coupled equations of motion of the linear aeroelastic model of wind turbines:

$$\begin{aligned} \mathbf{M}\ddot{\mathbf{u}} + (\mathbf{C} + \mathbf{G} + \mathbf{C}_a)\dot{\mathbf{u}} + (\mathbf{K} + \mathbf{K}_a)\mathbf{u} + \mathbf{A}_f \mathbf{x} &= \mathbf{0} \\ \dot{\mathbf{x}} + \mathbf{A}_d \mathbf{x} + \mathbf{C}_{ux} \dot{\mathbf{u}} + \mathbf{K}_{ux} \mathbf{u} &= \mathbf{0} \end{aligned} \quad (18)$$

These equations are autonomous enabling an eigenvalue analysis to determine the aeroelastic natural frequencies, damping, and mode shapes of wind turbines operating at a constant rotor speed.

## Eigenvalue Analysis

The aeroelastic equations of motion (18) couple the structural motion of the turbine with the dynamics of the aerodynamic forces. The structural dynamics are characterized by  $N$  natural modes with frequencies and

damping through a second order differential equation, whereas the dynamics described by the independent first order differential equation for each aerodynamic calculation point can be characterized as over-damped aerodynamic states.

The coupling of dynamic and purely damped states means that a direct formulation of an eigenvalue problem from (18) will lead to an ill-conditioned problem. This issue cannot be solved but the problem can be better conditioned by a modal expansion of the structural DOFs. The undamped eigenvectors of the stationary turbine (computed from the mass and stiffness matrix of the turbine at standstill)  $\mathbf{v}_i$  are used for this expansion:

$$\mathbf{u} = \sum_{i=1}^N \mathbf{v}_i z_i(t) = \Phi \mathbf{z} \quad (19)$$

where  $\Phi$  is a modal matrix containing the undamped eigenvectors in the columns, and  $\mathbf{z}$  is a new structural state vector.

To set up the eigenvalue problem all aeroelastic states are collected in a single state vector

$$\mathbf{y} = \begin{Bmatrix} \mathbf{x} \\ \mathbf{z} \\ \dot{\mathbf{z}} \end{Bmatrix} \quad (20)$$

and a solution of the form  $\mathbf{y} = \mathbf{w}e^{\lambda t}$  is sought. After use of expansion (19) and insertion of this solution form, the following eigenvalue problem is obtained

$$\left( \begin{bmatrix} \mathbf{A}_d & \mathbf{K}_{ux}\Phi & \mathbf{C}_{ux}\Phi \\ \mathbf{0} & \mathbf{0} & -\mathbf{I} \\ \Phi^T \mathbf{A}_f & \Phi^T (\mathbf{K} + \mathbf{K}_a)\Phi & \Phi^T (\mathbf{C} + \mathbf{G} + \mathbf{C}_a)\Phi \end{bmatrix} + \lambda \mathbf{I} \right) \mathbf{w} = \mathbf{0} \quad (21)$$

The eigenvalues  $\lambda_n$  and eigenvectors  $\mathbf{w}_n$  of this problem yields the natural frequencies, damping, and mode shapes of the aeroelastic turbine modes. The natural frequency and logarithmic decrement for mode number  $n$  are given by the imaginary and real part of the eigenvalues

$$f_n = \frac{\text{Im}\{\lambda_n\}}{2\pi} \quad \text{and} \quad \delta_n = -\frac{\text{Re}\{\lambda_n\}}{f_k} \quad (22)$$

and the corresponding mode shape can be obtained in physical (blade) coordinates from the eigenvector  $\mathbf{w}_n$  by the modal expansion (19) and the multi-blade transformation (1) for the blade coordinates. Identification of the modes can be done by animating the mode shapes using 3D graphics, or by computing the modal amplitudes for symmetric and whirling rotor motion as described in [5].

## Prediction Compared to Observation

Thomsen *et al.* [11] have developed an experimental tool for estimating the aeroelastic damping of turbine modes while the turbine is operating. In their experiment with a stall-regulated 600 kW Bonus turbine the low damping of the two edgewise whirling modes were estimated. The results showed that the forward edgewise whirling mode is more damped than the backward edgewise whirling mode, although they are associated with the same edgewise blade mode.

Hansen [5] presented a theoretical modal analysis of the particular turbine which showed that the blades vibrated more out of the rotor plane in the forward whirling than in the backward whirling mode. This difference can explain the measured difference in aerodynamic damping of the two modes because out of plane blade vibrations are more damped than in plane vibrations.

In this section the aeroelastic modes are computed with HAWCStab for the same 600 kW turbine. The results support the conclusion by Hansen [5], and also shows that the predicted aeroelastic damping of the two turbine



Table 1. Modelled and measured natural frequencies of the lowest 10 modes of the stationary turbine

| Mode no. | Name                                | Modelled freq. [Hz] | Measured freq. [Hz] |
|----------|-------------------------------------|---------------------|---------------------|
| 1        | 1 <sup>st</sup> shaft torsion       | 0.55                | 0.56                |
| 2        | 1 <sup>st</sup> long. tower bending | 0.76                | 0.73                |
| 3        | 1 <sup>st</sup> lat. tower bending  | 0.81                | 0.80                |
| 4        | 1 <sup>st</sup> yawing flap         | 1.41                | 1.2                 |
| 5        | 1 <sup>st</sup> tilting flap        | 1.50                | 1.5                 |
| 6        | 1 <sup>st</sup> symmetric flap      | 1.70                | 1.76                |
| 7        | 1 <sup>st</sup> vertical edgewise   | 2.92                |                     |
| 8        | 1 <sup>st</sup> horizontal edgewise | 2.99                | 2.9                 |
| 9        | 2 <sup>nd</sup> yawing flap         | 3.43                |                     |
| 10       | 2 <sup>nd</sup> tilting flap        | 3.64                | 3.6                 |

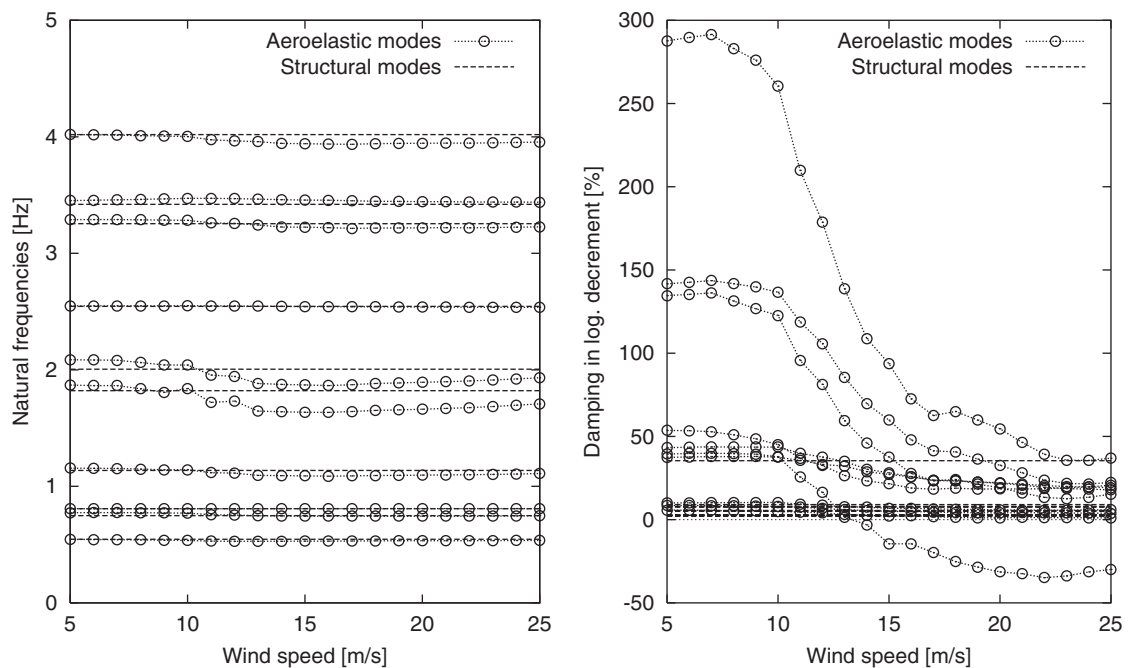


Figure 3. Natural frequencies (left) and damping (right) for the 10 lowest structural and aeroelastic modes of the Bonus 600 kW, predicted with HAWCStab for different operation wind speeds

modes corresponds well with the observations. The present analysis is very brief and it is recommended that the reader compares the present results with the more detailed study of the Bonus 600 kW turbine in [5].

The input to aeroelastic model is taken from the original HAWC model of the turbine used in the preparations for the experiment by Thomsen *et al.* [11], except for an improved calibration of structural stiffnesses to give the measured frequencies of the lower structural modes at standstill. These modelled and measured frequencies of the stationary turbine are listed in Table 1. Note that there has only been estimated a single frequency for each of the higher asymmetric mode pairs.

Figure 3 shows the structural and aeroelastic natural frequencies and damping of the 10 lowest modes of the turbine operating at different wind speeds. It shows that the natural frequencies are not affected signifi-

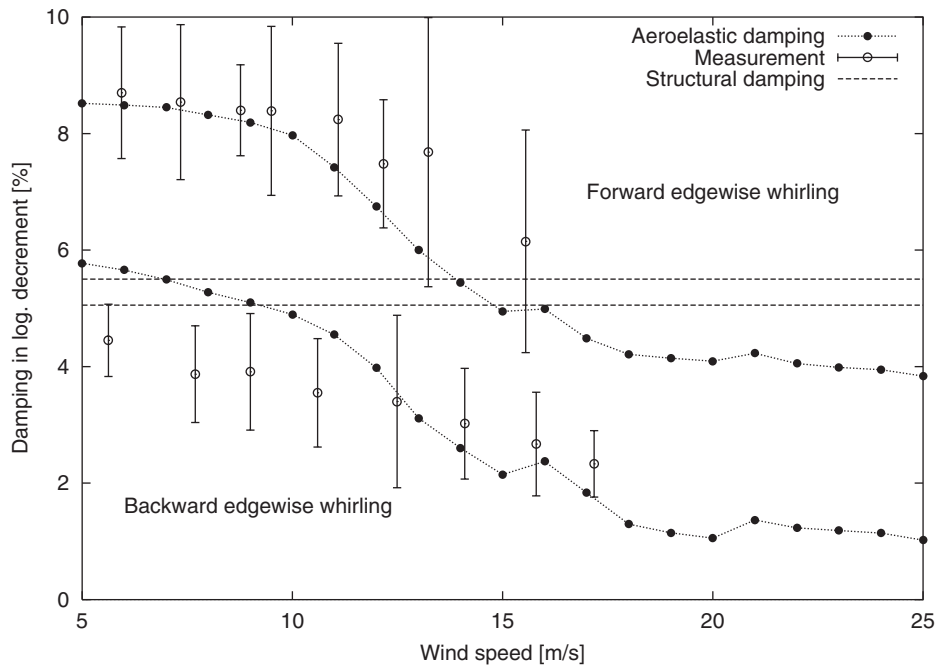


Figure 4. Predicted and measured aeroelastic damping of the forward and backward edgewise whirling modes of the Bonus 600 kW turbine, together with the structural damping to show the change in damping due to the interaction with aerodynamic forces

cantly by the aerodynamic forces, except for the modes involving flapwise blade modes, which are also the most highly damped modes seen in the damping diagram. The mode that becomes negatively damped as the wind speed increases is the symmetric edgewise (first shaft torsion) mode because of the blade vibrations purely in the rotor plane. On the actual turbine, this mode will be damped by the drive-train damping due to the slip of the induction generator.

The two lowest damped modes are the two asymmetric edgewise modes, i.e., the forward and backward edgewise whirling modes. Figure 4 shows the predicted and measured aeroelastic damping of these two modes, together with the structural damping for comparison. Although the standard deviations of the measured damping is relatively large, it is clear that these modes have different aeroelastic damping. The same qualitative difference is seen in the predicted aeroelastic damping, and furthermore the level of damping is very similar to the observed damping. These first results with the new tool for stability analysis indicate that it is possible to predict the qualitative and quantitative behavior of the damping of aeroelastic turbine modes.

## Conclusion

In this paper a design tool, called HAWCStab, for performing aeroelastic stability analysis of wind turbines is presented. The aeroelastic turbine model is derived from a Finite beam Element Method and the Blade Element Momentum method, where the unsteady aerodynamic forces are modelled by a Beddoes-Leishman type dynamic stall model in a state-space formulation. Multi-blade coordinates are used directly in the derivation of the linear aeroelastic equations of motion to eliminate the periodic coefficients and avoid using the Floquet Theory. The eigenvalue problem based on these autonomous equations is ill-conditioned, however its condi-

tion is improved by using a modal expansion of the structural degrees of freedom based on the undamped modes of the turbine at standstill. This enables the computation of the natural frequencies, logarithmic decrements, and mode shapes of the aeroelastic turbine modes. An example shows a good agreement between predicted and measured aeroelastic damping of a stall-regulated Bonus 600 kW turbine.

## Acknowledgements

This work was partly funded by the European Commission under ENK5-CT-2002-00627 which is gratefully acknowledged.

## References

1. Kirchgäßner B. ARLIS—A program system for aeroelastic analysis of rotating linear systems. In W. Palz, editor, *Proceedings of the European Wind Energy Conference 1984*, pages 253–264, Hamburg, Germany, October 1984.
2. Stiesdal H. Extreme wind loads on stall regulated wind turbines. In G. Elliot, editor, *Proceedings of the 16th British Wind Energy Association Conference*, pages 101–106, June 1994.
3. Petersen JT, Madsen HAa, Björck A, Enevoldsen P, Øye S, Ganander H, Winkelaar D. Prediction of dynamic loads and induced vibrations in stall. Technical Report Risø-R-1045(EN), Risø, Roskilde, Denmark, May 1998.
4. Rasmussen F, Petersen JT, Madsen HAa. Dynamic stall and aerodynamic damping. *ASME Journal of Solar Energy Engineering*, 1999; **121**: 150–155.
5. Hansen MH. Improved modal dynamics of wind turbines to avoid stall-induced vibrations. *Wind Energy*, 2003; **6**: 179–195.
6. Chaviaropoulos P, Politis ES, Sørensen NN, Hansen MH, Bulder BH, Winkelaar D, Saravanos DA, Philippidis T, Galiotis C, Hansen MO, Kossivas T. Recent advances on damped wind turbine rotor blades, the DAMPBLADE project. In *Proceedings of the European Wind Energy Conference 2003*, Madrid, Spain, June 2003. (to appear).
7. Chaviaropoulos PK. Flap/lead-lag aeroelastic stability of wind turbine blades. *Wind Energy*, 2001; **4**(4): 183–200.
8. Petersen JT. *Kinematically Nonlinear Finite Element Model of a Horizontal Axis Wind Turbine*. PhD thesis, Risø National Laboratory, DK-4000 Roskilde, Denmark, 1990.
9. Petersen JT. The aeroelastic code HawC—model and comparisons. In Pedersen BM, editor, *State of the Art of Aeroelastic Codes for Wind Turbine Calculations*, volume Annex XI, pages 129–135, Lyngby, April 1996. International Energy Agency, Technical University of Denmark.
10. Hansen MH, Gaunaa M, Madsen HAa. A Beddoes-Leishman type dynamic stall model in state-space and indicial formulations. Technical Report Risø-R-1354(EN), Risø National Laboratory, 2004. (to appear).
11. Thomsen K, Petersen JT, Nim E, Øye S, Petersen B. A method for determination of damping for edgewise blade vibrations. *Wind Energy*, 2001; **3**: 233–246.
12. Johnson W. *Helicopter Theory*. Princeton University Press, New Jersey, 1980.
13. Peters DA. Fast floquet theory and trim for multi-bladed rotorcraft. *Journal of the American Helicopter Society*, 1994; **39**(4): 82–89.
14. Bazoune A, Khulief YA. Shape functions of three-dimensional Timoshenko beam element. *Journal of Sound and Vibration*, 2003; **259**(2): 473–480.
15. Hansen MH. Anisotropic damping of Timoshenko beam elements. Technical Report Risø-R-1267(EN), Risø National Laboratory, 2001.
16. Meirovitch L. *Methods of Analytical Dynamics*. McGraw-Hill, 1970.